Loadscope User Guide

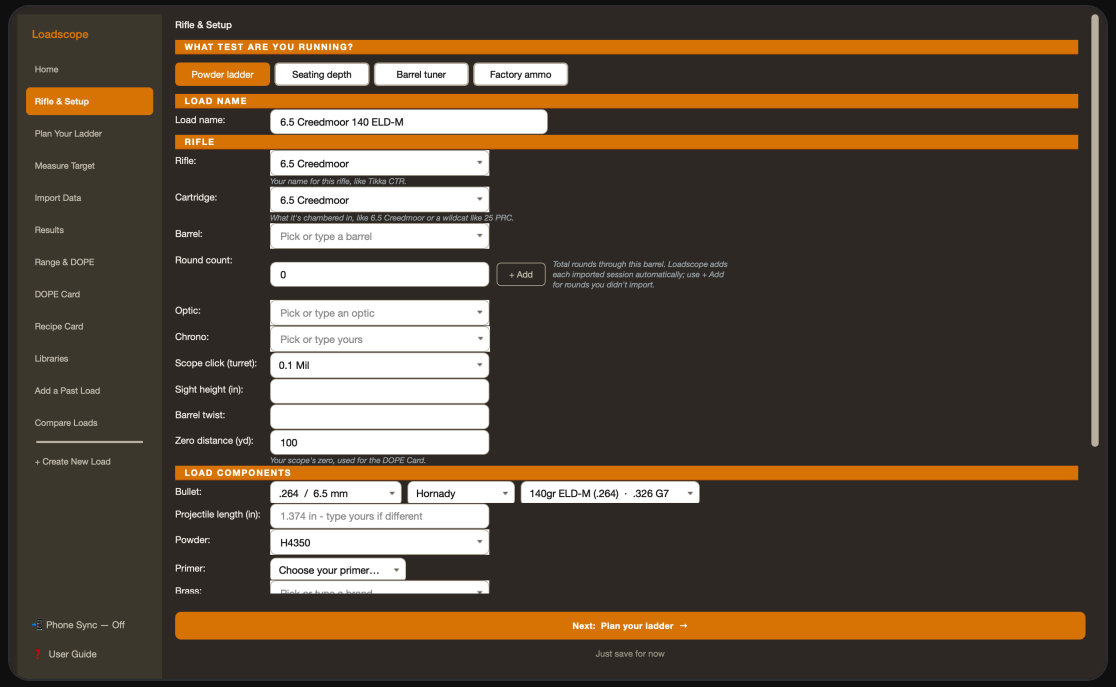
Precision rifle load development for Mac & Windows

Version 1.17.4 · loadscope.app

ONCE PER RIFLE

Create New Load — set up your rifle

Click Create New Load in the left sidebar. Step 1 is your rifle and setup. Loadscope writes it into your workbook for you — no typing into spreadsheet cells.



1. Load a saved setup — pick a previous setup from this dropdown to fill everything in, or start a new one by filling in the fields below. Loadscope saves what you enter, so it's in this dropdown next time.

2. What test are you running? — Powder ladder, Seating depth, Barrel tuner or Factory ammo. Pick it first: it decides which boxes the rest of the form shows.

3. Rifle — rifle, cartridge, barrel, optic and scope click. Set once; pick it from the dropdown next time.

4. Zero distance — the range your scope is zeroed at. The DOPE Card and Range & DOPE work from it, and it carries over with the rifle to your next load. It is not the same as Test distance, which is how far you shot this ladder.

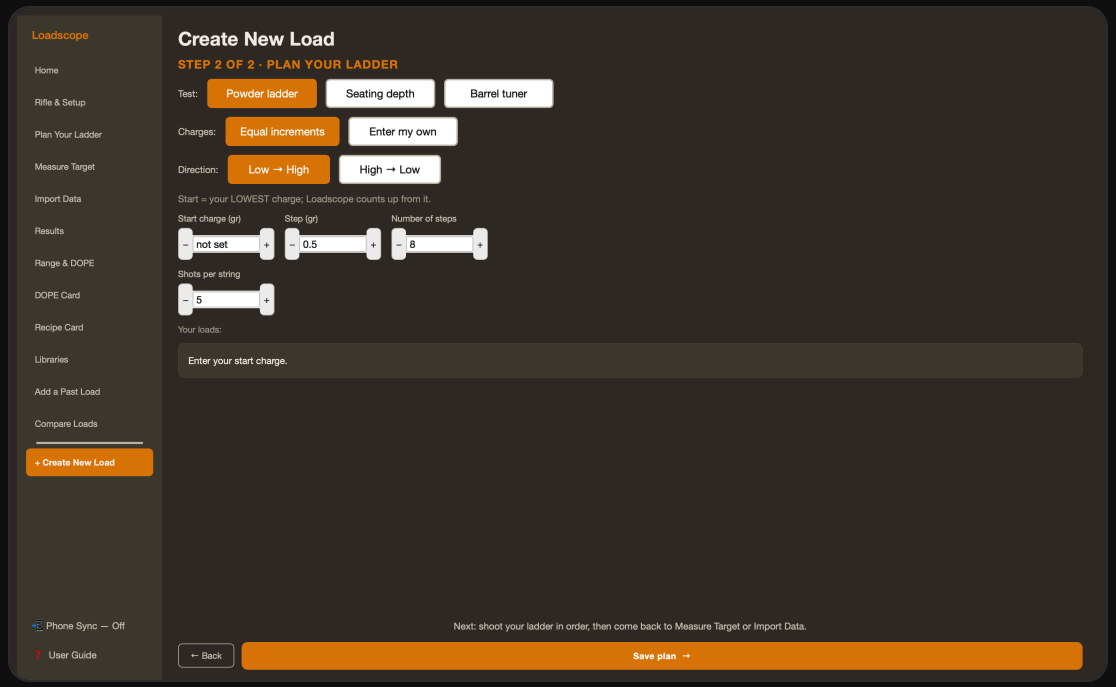
5. Load Components — bullet, powder, primer, brass. The bullet picker also feeds the ballistic solver.

6. Projectile length — optional, and you can usually leave it alone. Picking your bullet fills this in from the maker's published figure where there is one, and Loadscope estimates it otherwise. The box tells you which you are getting. Type a number only if you have measured the bullets in front of you with calipers — makers publish an approximate length and it can differ from the lot you actually bought, so your own measurement wins. This is the length of the bullet on its own, not a loaded round.

7. Click Next: Plan your ladder → to move on.

Plan your ladder

Enter your ladder one time and Loadscope lists every load to make.



> Then shoot them in order — lowest charge to highest, or highest to lowest if you set the plan to High → Low. Shooting in order is how Loadscope matches each string to the right charge. This is the key to the whole process.

The planner is also on the left sidebar: Plan Your Ladder opens it for the load you are on.

Record by hand at the range? Tools → Print Ladder Sheet prints your planned charges in shooting order, with blank spaces for speeds and conditions (powder and seating-depth ladders).

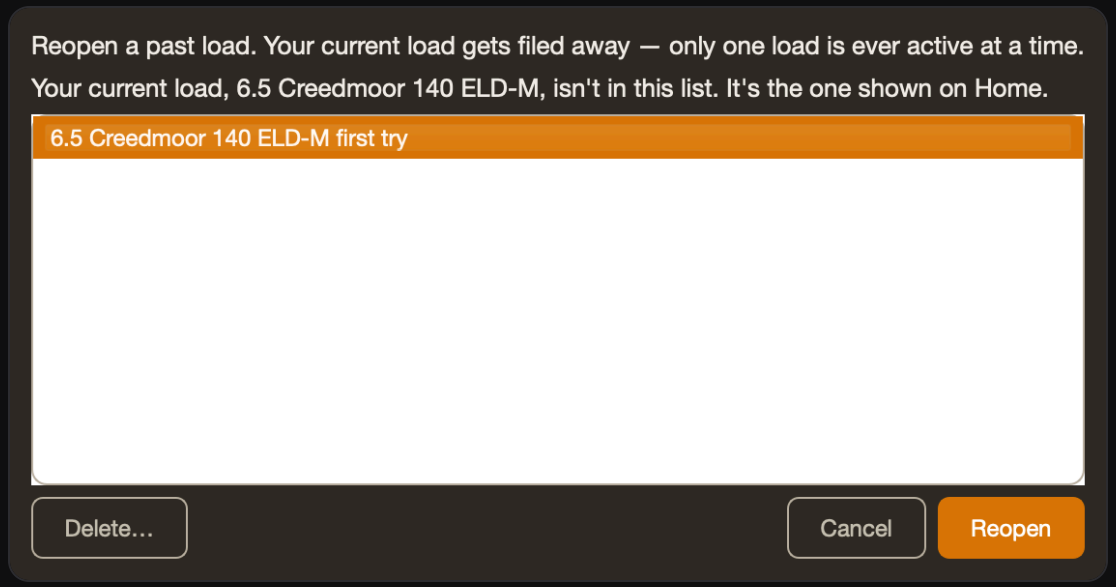
1. Powder ladder, Seating depth or Barrel tuner — pick which test this is. A powder ladder steps the charge; a seating test steps the seating depth, entered as a CBTO (2.780 in), a COAL (overall length, if you don't use a comparator) or a jump off the lands (0.020 in) — whichever way you measure; a tuner test steps the setting on your tuner dial and leaves your load alone.

2. Enter it once: your starting charge, the step between charges, and how many. Loadscope lists every load below so you know exactly what to load. Equal increments steps them evenly; Enter my own lets you type each one.

LOADS YOU HAVE FILED AWAY

Your earlier loads

Want an earlier load back? Click Past Loads… on the Loads tab in Libraries. It lists every load you have filed away, including ones you never saved to the Library. To remove one, pick it and click Delete… — it goes to the Trash, so you can get it back.



Deleting a load from the Library removes only its saved recipe; the test it came from stays in Past Loads. And a different test always gets its own load: if you import new charges into a load that already has speeds, Loadscope steers you to a new load, so two tests are never scored together. Choose Add to this test only when you went back and shot the same charges again. Adding your chronograph or target data to a load you already started goes straight in with no question.

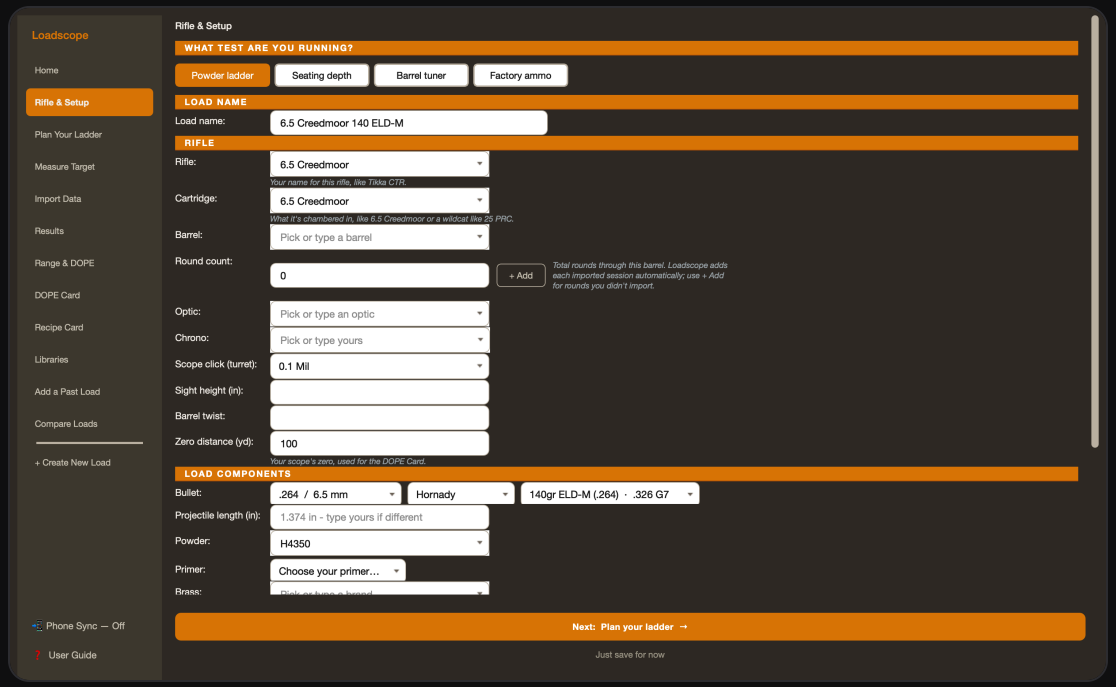
Worked up a load before you had Loadscope? Click Add a Past Load in the left sidebar (under Libraries) and type it in: the rifle, the recipe, and whatever results you kept (velocity, SD, extreme spread, group size and the distance you shot it at). It is filed in Past Loads for your records and never replaces the load you are working on.

Reworking a load? Use Rework a Saved Load on the Home screen — it opens here with that load's rifle already filled in.

TRACK BARREL LIFE

Track rounds per barrel

Loadscope keeps a running round count for every barrel — set it once and it stays current on its own.



Round count lives right on Rifle & Setup, tied to the barrel you entered.

Set a starting number once — every import adds that session's rounds automatically.

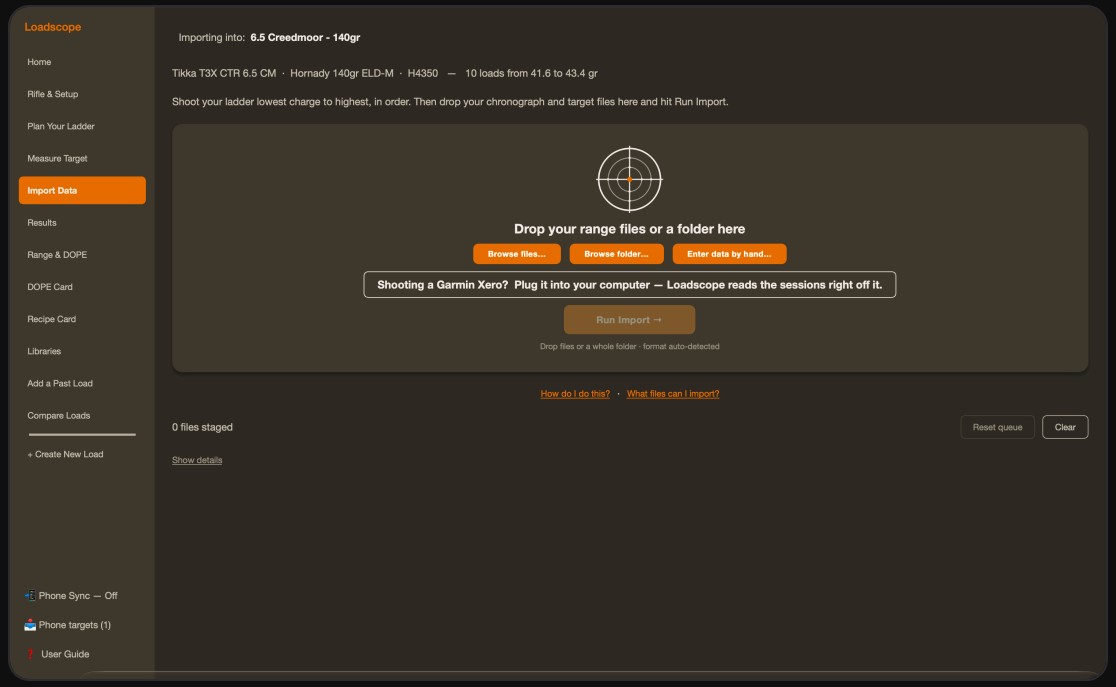
Shot rounds you didn't import (practice, matches)? Click + Add and type how many.

Switch barrels and the count follows — each barrel keeps its own total.

EVERY RANGE DAY

Shoot, then drop your files

Back from the range? Drop the files your chronograph and target app export onto the drop zone — or click Browse — then Run Import.



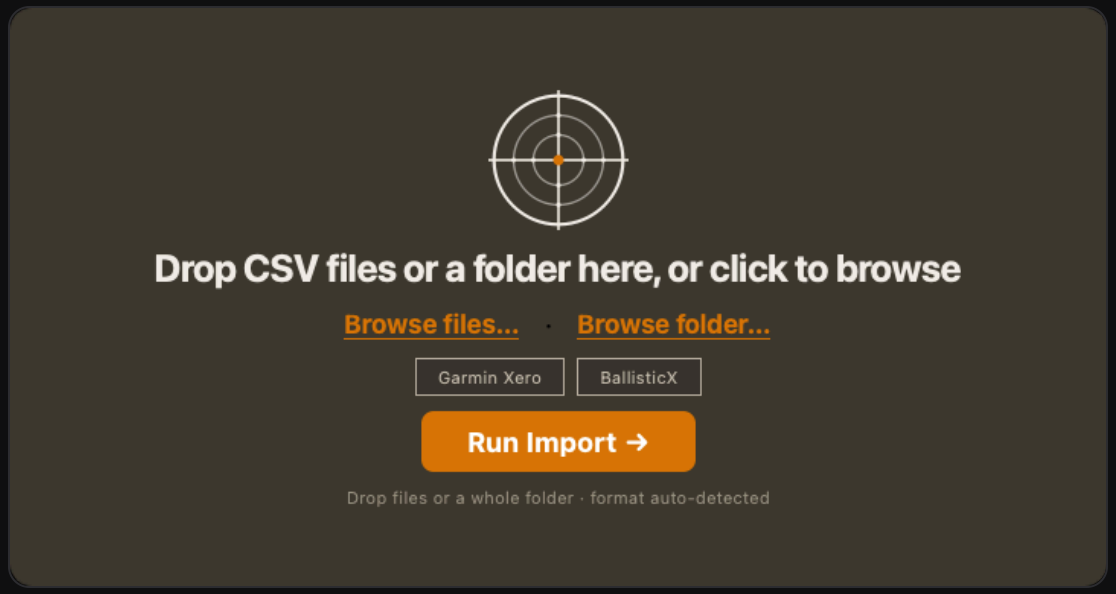
The line at the top, Importing into:, names the load your files go into. Anything that needs your attention shows right on the page; Show details opens the full list of what the import did.

1. Drop zone — bring in your chronograph file (and a target app's export too, if you use one). The Chronograph and Target app chips show where each kind goes.
2. Click Run Import. Loadscope backs up your workbook first, matches everything to your ladder, and shows your results right inside the app — click See your results → beside Run Import to go straight there.

FASTER IMPORTS

Import a whole folder

Point Loadscope at a whole range-day folder and it imports everything inside — no file-by-file dragging.



Drag a whole range-day folder onto Loadscope, or click Browse folder.

It pulls every chronograph file inside — and your target app's exports too, if you use one — including subfolders, in one shot. It doesn't care what the files are called or what format they're in.

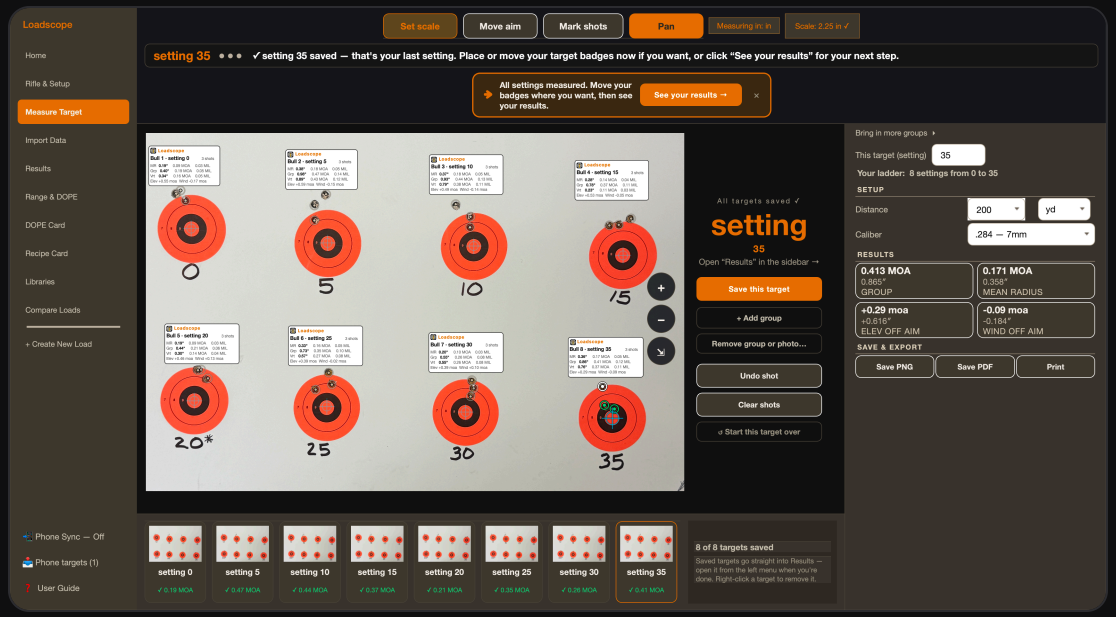
Each file's format is auto-detected, so you never sort them by hand.

Individual files still work the same way — point it at a folder when that's faster.

NO TARGET APP NEEDED

Measure your target groups from a photo

Don't run a target app? The Measure Target page measures your group right in Loadscope and scores it like any import.



1. Bring in your groups — Import from phone pulls groups you measured on the Loadscope Ballistics app over Wi-Fi; Choose target photos lets you measure on the desktop. Until a photo is loaded the page shows only these buttons.

2. Set the scale — drag a line across a known length (a 1-inch grid square) and type what it measures. Work in inches, mm, MOA or mil, whichever your target is printed in — many shooters run a mil-dot scope and a mil grid. That turns pixels into real measurements, so every number is honest.

3. Set your aiming point — click Move aim and drop the crosshair on your point of aim. Loadscope then measures how far the group sits off that aim — elevation and windage — not just its size.

4. Mark shots — tap each bullet hole. Loadscope works out group size, mean radius and vertical, and scores it under its charge.

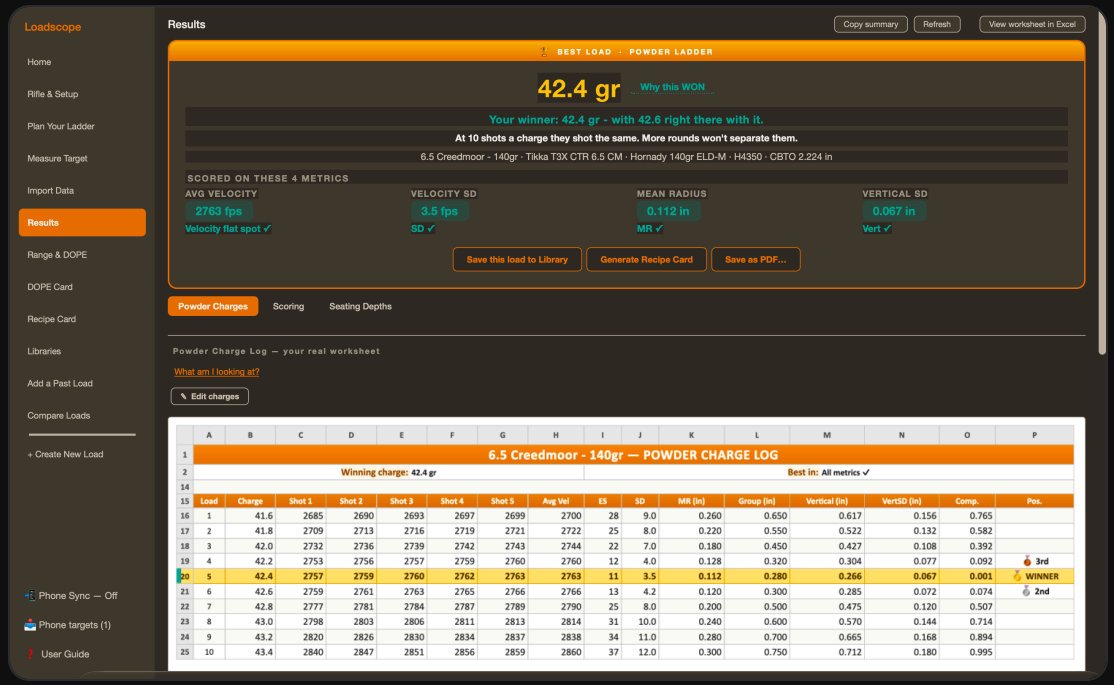
5. Save this target — when the last shot is marked, the bar above the photo tells you to press Save this target. Nothing is saved until you press it. When every group sits on its own photo it stays put afterwards, so you can drag the result and recipe boxes clear of your group before you move on; the button you press next is the one that's lit.

6. The test and the distance come from Rifle & Setup — the line above Results shows the test, the test distance and the bullet size, with a link to change them. A target will not save until a test distance is set. Measure a whole ladder on one sheet — a bull per charge, added with + Add group — in one pass.

THE ANSWER

Your best powder charge

The Results screen leads with your winner. The gold BEST LOAD bar at the top is the suggested charge — picked from the loads you tested. Three per-winner buttons sit right under the numbers.



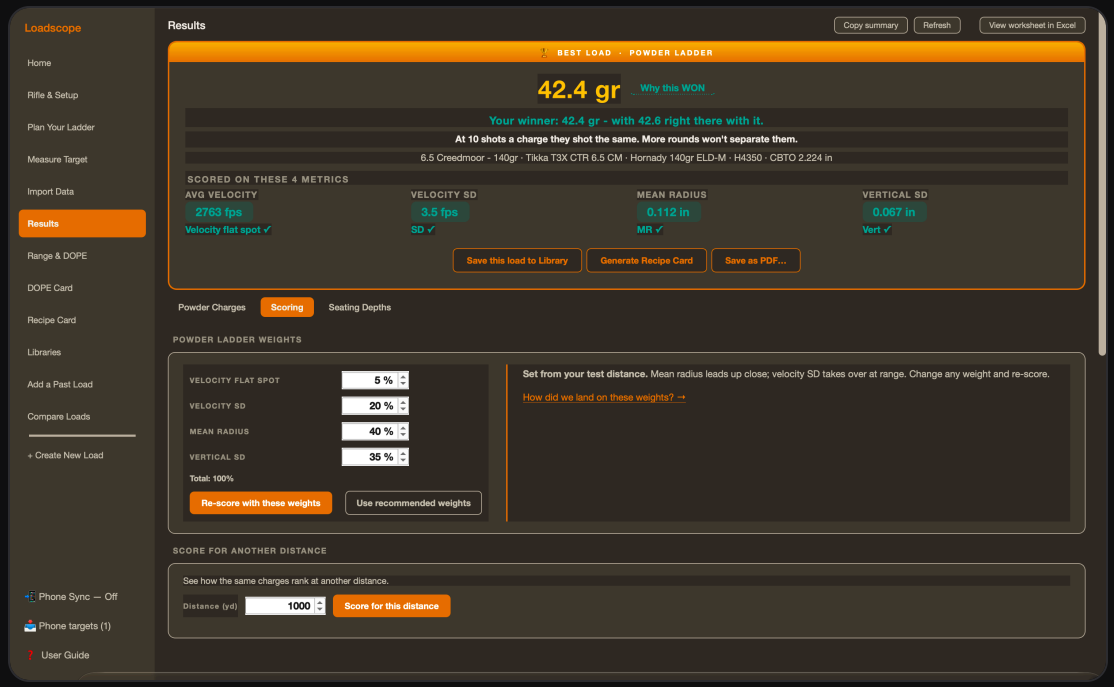
Three buttons under the numbers — Save this load to Library, Generate Recipe Card (printable recipe in your browser), and Save as PDF.

1. BEST LOAD — the winning charge weight, scored across velocity flat-spot, SD, mean radius and vertical SD. Need at least 3 charges (plan 5–8 in a ladder).
2. Powder Charges — your real worksheet, right in the app. Composite Score: lower is better.
3. Every shot — the button above the worksheet lists every shot's speed for each charge, however many you fired. The worksheet itself shows the first five.
4. Winner column — 1st / 2nd / 3rd ranking so you can see the runners-up.

HOW THE SCORE IS BUILT

Scoring — why this charge won

The Scoring sub-tab shows the scoring math — exactly why a charge won, and lets you tune what matters to you.



Four things are scored: velocity flat-spot, velocity SD, mean radius and vertical SD. Each is put on the same scale, then blended into one composite score — lowest wins.

> The four are NOT weighted equally, and the mix is not a fixed table — Loadscope sets it from the distance you tested at. Up close, mean radius leads and velocity barely counts. At distance, velocity SD takes over, because that is what opens your groups up vertically the further you shoot.

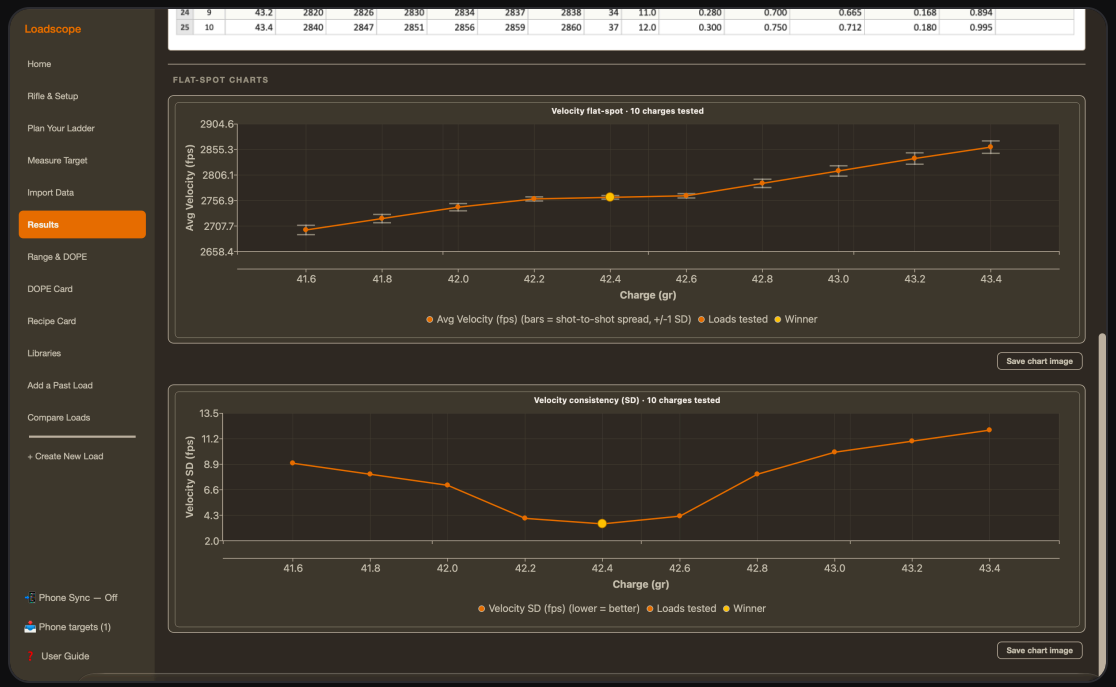
A powder ladder at 100 yards comes out at mean radius 40%, vertical SD 35%, velocity SD 20% and flat-spot 5%. By 1000 yards velocity SD has climbed to 50% and mean radius has dropped to 25%. A seating test is weighted differently again: velocity barely counts at all (10%), mean radius carries it up close at 65%, and vertical SD grows to 40% by 1000 yards.

Tune what matters to you — every weight is yours to set on the Scoring tab. Nudge one and re-score. Use recommended weights hands the dial back to Loadscope.

READ THE FLAT-SPOT

Charts with ±1 SD error bars

The charts live on the sub-tab where the data is: open Powder Charges for the velocity and velocity-SD charts, Seating Depths for the mean-radius and vertical-SD charts. Every charge is graphed with ±1 SD error bars so your velocity flat-spot is obvious at a glance.



Every charge is plotted with ±1 SD error bars on each point.

A flat run of points with short whiskers is your velocity flat-spot — it pops right out.

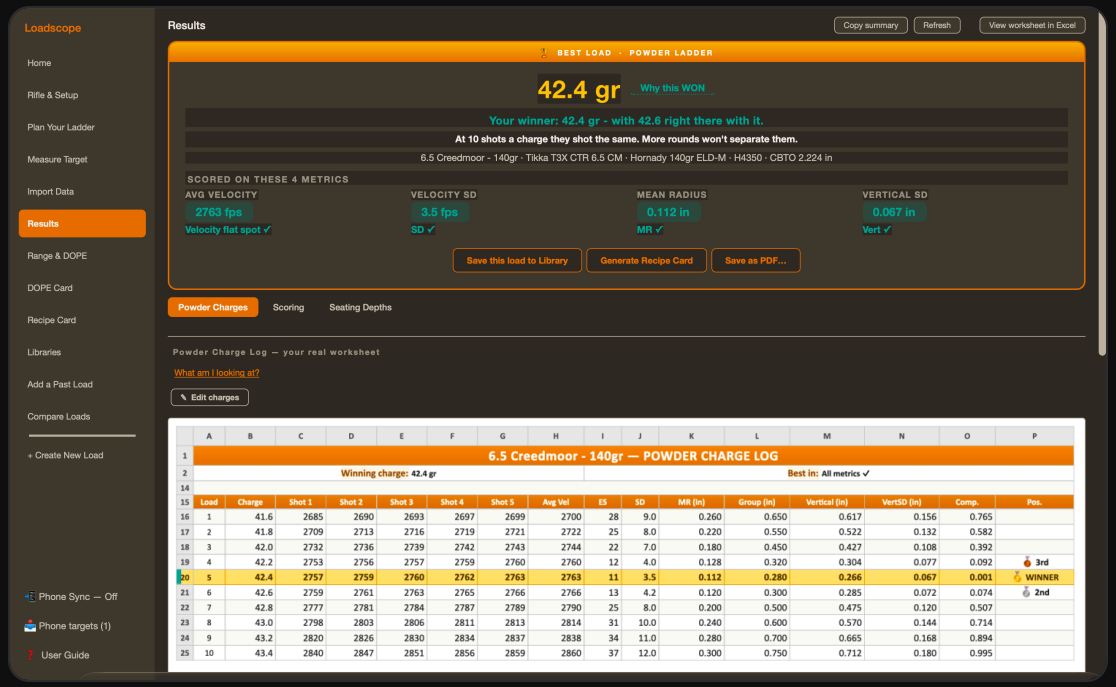
The winning charge is highlighted on the curve.

Hit Save chart image for a clean PNG to keep or share.

THE SEATING ANSWER

Your best seating depth

The Seating Depth view — same logic, but the bar at the top picks the winning seating depth instead of the winning charge.



1. BEST CBTO — the winning seating depth, scored across the same four factors. Shown the way you entered the ladder: a CBTO (base to bullet ogive), a COAL (overall length) or a jump off the lands. Plan at least 3; 5–7 is plenty.

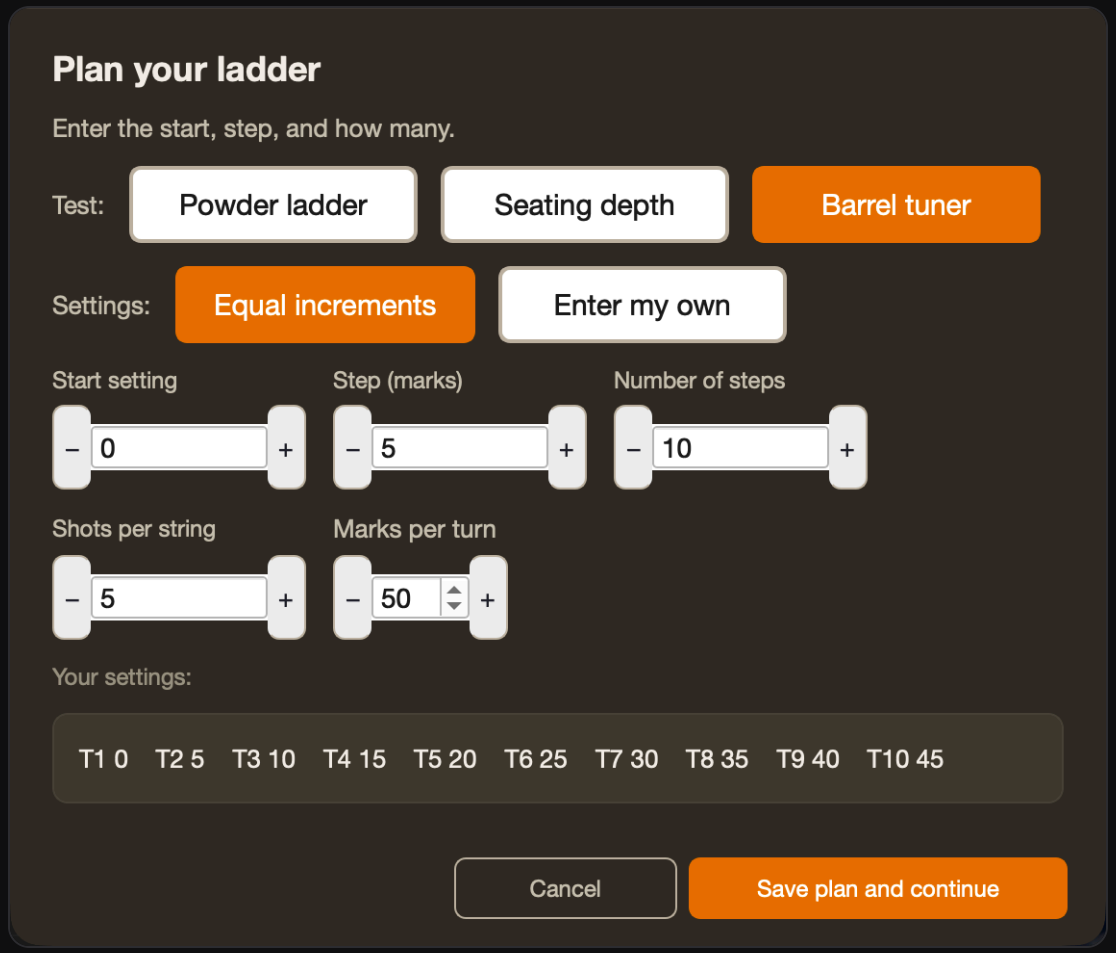
2. CBTO (in) — each row's tested seating depth. Loadscope fills these from your seating-depth strings, matched in order by your ladder plan (nothing to name).

3. Winner column — 1st / 2nd / 3rd by composite score. Save to Library to keep your confirmed load.

THE TUNER ANSWER

Your best tuner setting

Running a barrel tuner? Loadscope scores each setting you shoot on vertical spread and mean radius, then names your best setting and the run of good settings beside it — or the mark between two good settings, as the tuner maker advises. Shoot 5 shots a setting if you can. Your powder charge and seating depth stay where they are — only the dial changes.



Pick Barrel tuner on the Plan your ladder screen, then enter your start setting, the step in marks, and how many settings you want to shoot. Follow your tuner maker's instructions for how far to move it; Loadscope always counts up from your start setting. Enter Marks per turn as well — how many marks make one full turn of your tuner, counted on your tuner's dial. That is what tells Loadscope the last setting on the dial sits next to the first.

Shoot one group at each setting, in the order listed, and measure them the way you measure any target. Loadscope pairs each group to its setting by position, so there is nothing to name.

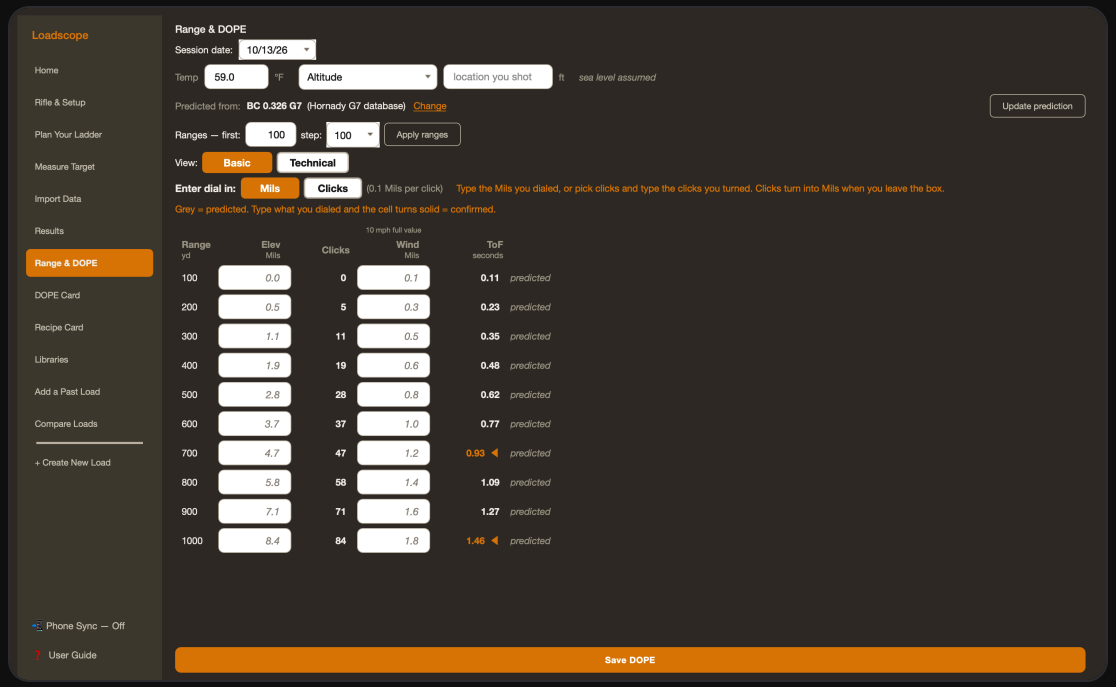
> Velocity is deliberately left out of a tuner score. It barely moves with tuner position, so scoring it would be measuring your ammunition and calling it a tuner result.

Settings too close together to tell apart are reported as too close to call rather than ranked — three-shot groups will do that often. You set how the score splits between vertical spread and mean radius on the Scoring screen.

BEFORE YOU SHOOT

Range & DOPE

From your winning load and the measured muzzle velocity, Loadscope predicts your full DOPE table — 100 to 1000 yards.



1. Predicted from your bullet's BC and drag model (G7/G1 solver, checked against published Applied Ballistics figures). Set the temperature and the altitude where you shot — or station pressure, if you carry a weather meter — then click Update prediction. Change, beside Predicted from, lets you type your own BC.

2. Basic or Technical — Basic shows elevation, clicks, wind and time of flight. Technical adds spin drift, Coriolis and lead, and a Latitude box for Coriolis.

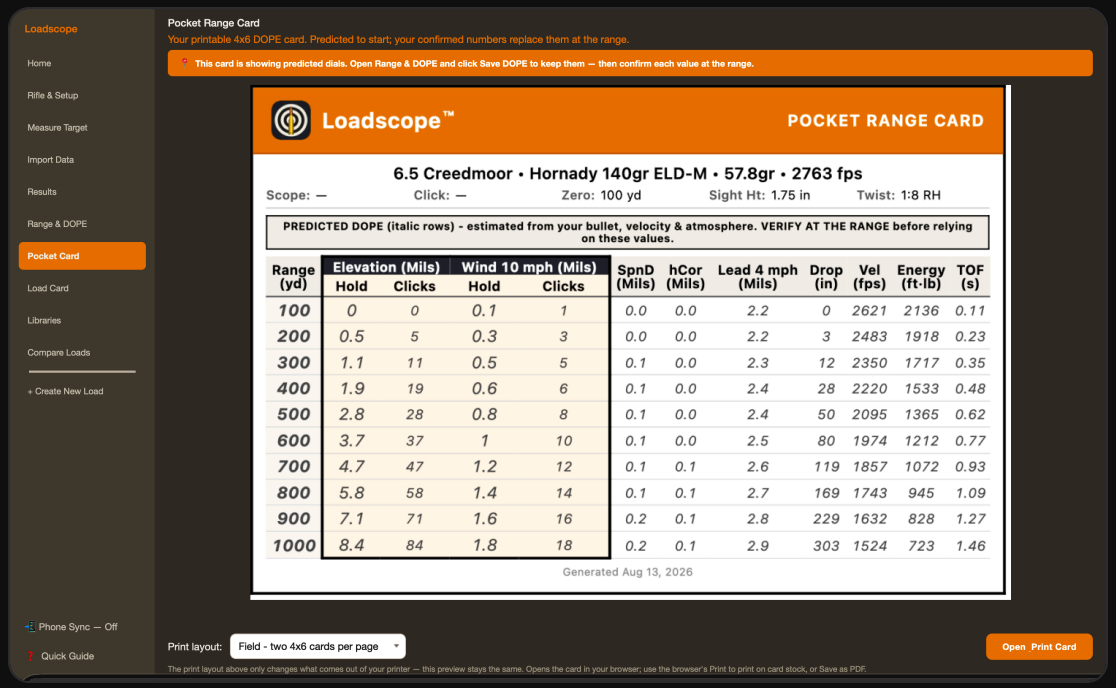
3. Grey, italic = predicted. Type what you actually dialed at the range and the cell turns solid = confirmed. Type two of them and Loadscope fills in the ranges between, and carries your correction a little further out — to 600 yards, no further. Past that it goes back to a plain prediction rather than guessing.

4. Click Save DOPE to write it back. Always confirm by live fire at known distance before you trust it.

TAKE IT WITH YOU

DOPE Card

A 4×6 range card you can print or save as PDF. Predicted to start; the numbers you dialed replace them as you shoot.

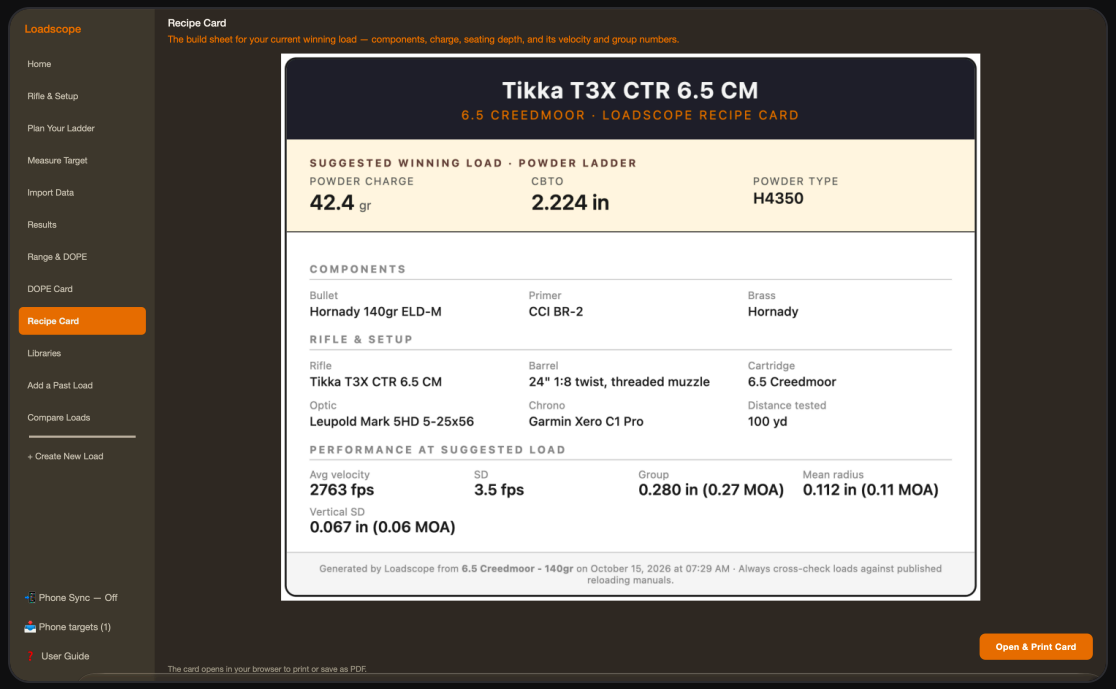


1. Header — the load, scope, zero, sight height, twist. Everything you need on a card glanced at in the field.
2. Dial table — elevation and clicks in your turret's own units (mil or MOA — Loadscope reads which from your scope-click setting), a fixed 10 mph wind reference, drop, velocity, energy, and time of flight per 100 yards. Print the 4×6 card or Save as PDF.
3. Whose number is it — grey italic is our prediction, solid is what you dialed, and a shaded row is worked out from two of your own. A line under the table says how many ranges you have confirmed and where the projection stops.

KEEP THE RECIPE

Print your Recipe Card

The Recipe Card is the one-page printable recipe you produce with the buttons on Results (Generate Recipe Card and Save as PDF…). Bullet, powder, charge, primer, brass, seating length (labelled CBTO or COAL, the way you measured it), plus the performance numbers that won — everything you'd need to load it again next season, on a single sheet.

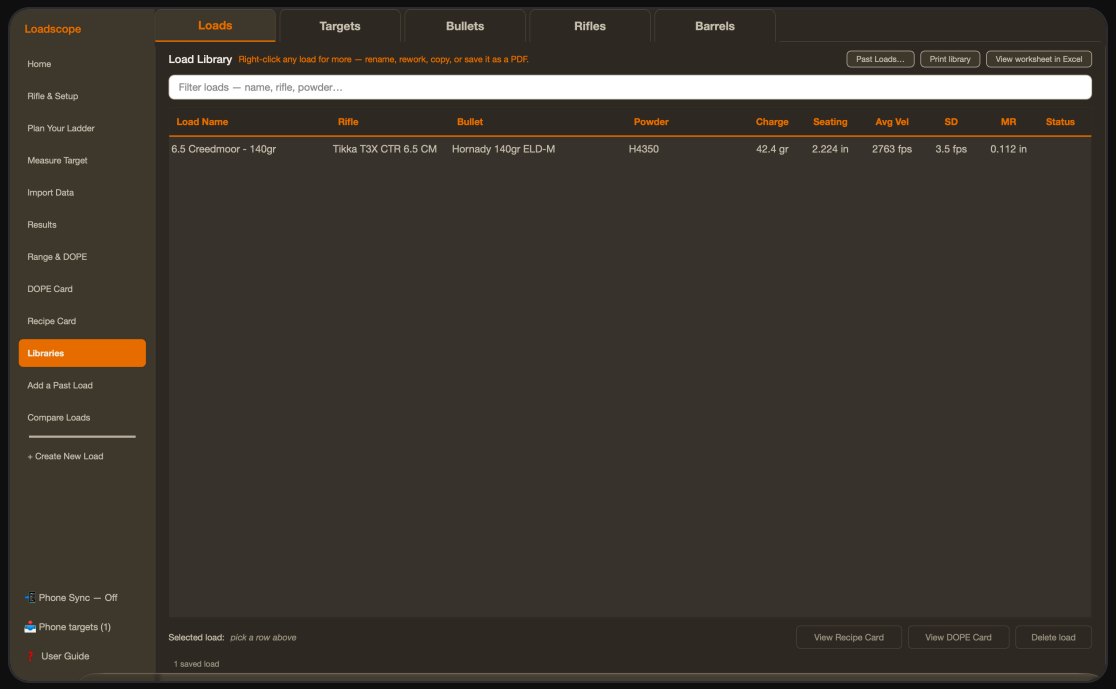


1. Print button — right on the page. One click sends the recipe to your printer for the binder on your bench.
2. Suggested winning load — the same charge, seating length and powder from your Results screen, restated for a printed header.
3. Components + Rifle setup — every part of the recipe and the rifle it ran on.
4. Performance at suggested load — the avg velocity, SD, group, and mean radius the winning load actually produced.
5. Card size — a full page, a 3x5 card for the ammo box, or 3x5 label sheets (Spartan R031 or Avery 94213) to stick on the box. Loadscope remembers your pick.

LOADS OVER TIME

Your Load Library

Every load you save with the Save this load to Library button ends up in your Load Library — across cartridges, rifles, and seasons. Right-click any row for the actions you'll use most.



The Library lives inside the same workbook as your Results — your saved loads travel with the file, with no extra cloud account to manage.

1. Saved loads — one row per load you confirmed. Click any row, then right-click for the actions menu.

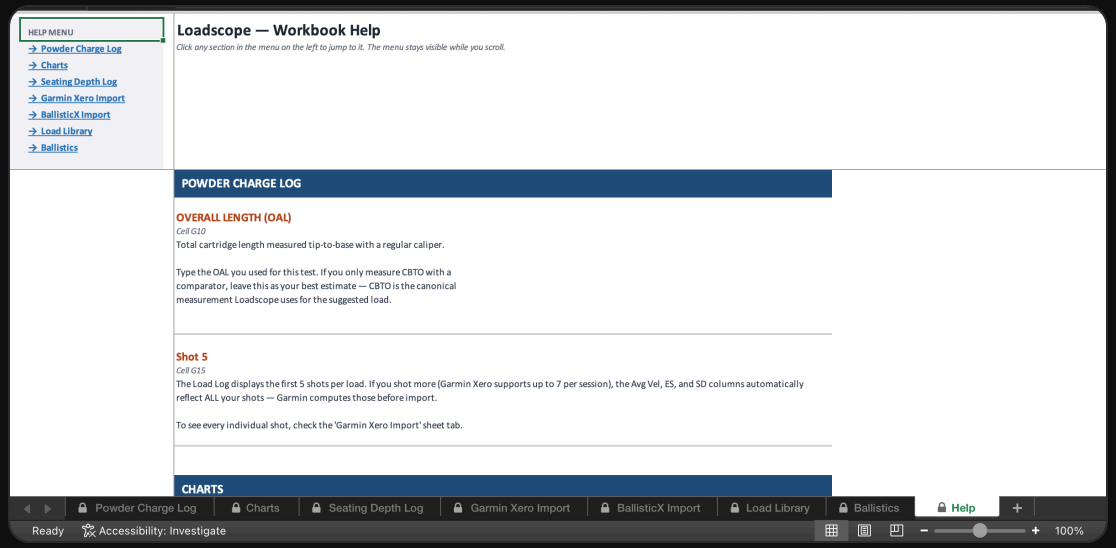
2. Grouped by rifle — loads sit under each rifle, newest first, with the date each one was saved. Click a rifle's heading to fold it shut. Double-click a load to open it on Rifle & Setup (right-click > Open this load does the same). Right-click a load to rename its rifle, and every load under that rifle follows.

3. Right-click actions — Copy load summary (paste into a text or email), Enter your actual dials… (reopens that load on Range & DOPE so you can type what you came up at each range, even on a load you already finished), Print this load as a Recipe Card… (same recipe as Results, but for a saved load), Save this load as PDF… (same native save dialog as Results), Open in Excel at this row (jump straight to the workbook), Delete this load… (confirm before anything goes).

IN-WORKBOOK HELP

The Help tab — column meanings & formulas

The Help tab lives at the right end of the bottom tab bar. It replaces the old cell hover-comments (which Excel-for-Mac was clipping). The full text is always visible.



When a label or number isn't self-explanatory, this is the place to look — no scrolling through PDFs, no hover-popup guesswork. It's part of the workbook itself.

1. HELP MENU sidebar — click any section name to jump straight to that tab's help. The menu stays visible while you scroll the body.
2. Section banner — names which workbook tab the items below are about (Powder Charge Log, Charts, Seating Depth Log, and so on).
3. Each item: bold header (the concept), the cell reference (e.g. Cell G15), and a plain-language explanation of what it means or how it's calculated.

TO YOUR PHONE

Send a load to your phone

Loadscope syncs straight to the Loadscope Ballistics app on your iPhone or iPad over your local Wi-Fi — no cables, no cloud, no account. Your dial goes where you actually shoot.



On the computer: turn on Phone Sync — the Phone Sync button at the bottom of the sidebar, or Support → Phone Sync. Your computer becomes discoverable on the Wi-Fi by name (it uses your Loadscope license, so only your own devices can connect).

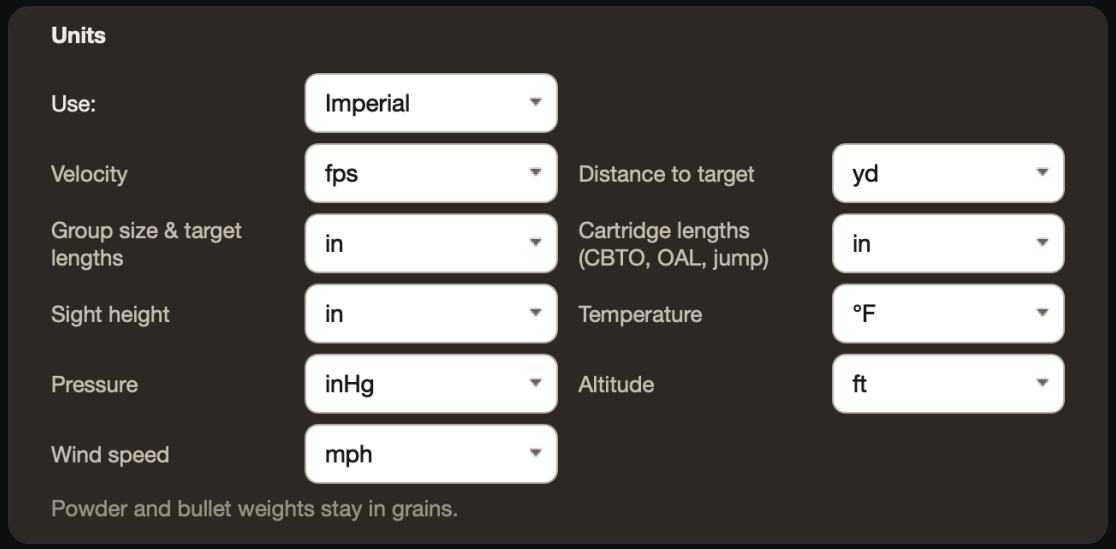
On the phone: open Loadscope Ballistics, tap Sync, pick your computer, and pull the load — the recipe and predicted DOPE land on your phone for the field. Keep both devices on the same Wi-Fi.

On a guest or corporate network that hides devices? Tap Connect by address on the phone and type the address the computer shows.

YOUR UNITS

Imperial or metric

Open Settings and find Units. Pick Imperial or Metric to set everything in one click, or choose each measurement on its own — velocity in m/s with group size in inches is fine. The choices cover velocity, distance to target, group size, cartridge lengths (CBTO, OAL, jump), sight height, temperature, pressure, altitude and wind. Powder and bullet weights stay in grains.



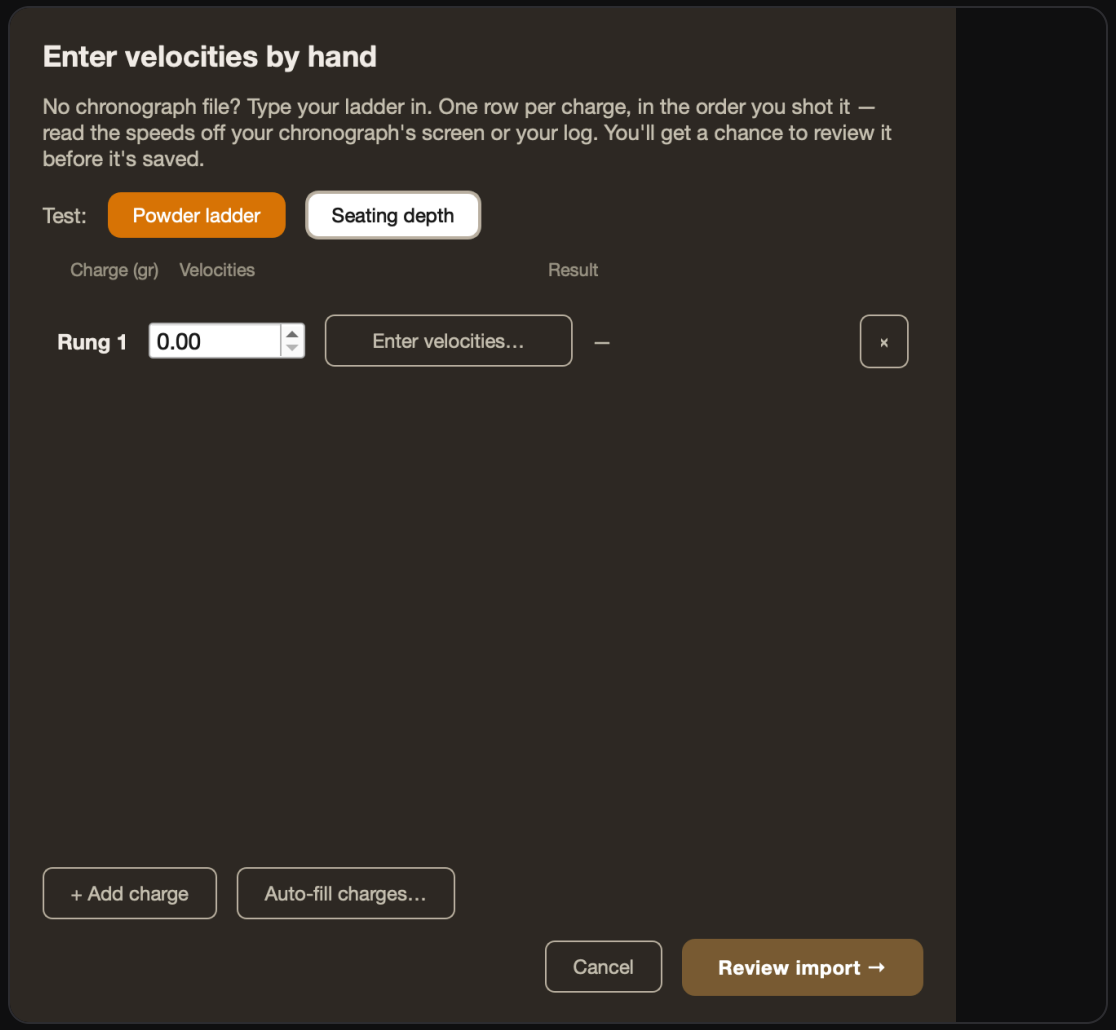
Units only change what you see and type. Your saved loads, your scores and your workbook are stored the same way whatever you pick, so you can switch back and forth freely. The Excel workbook itself stays in inches, yards and fps.

Files from a European-set computer import as they are: m/s from a Garmin, mm group sizes, decimal commas and semicolon-separated columns all read correctly.

OTHER SETUPS

If your gear isn't on the list

Loadscope reads Garmin Xero (plugged in over USB, or a ShotView export), LabRadar, MagnetoSpeed and Athlon Velocity Pro. For targets it reads ShotMarker, BallisticX, OnTarget and Silver Mountain — or you can skip the target app and measure your groups from a photo right in Loadscope.



> No file from your chronograph? Type the velocities in. On the Import screen, next to Browse folder, click Enter data by hand and put in the velocities for each charge in the order you shot them — straight off the chronograph's screen or your range log. They are scored exactly the same as an imported file.

Shooting factory ammo? Click Create New Load and set the test to Factory ammo, then type what's on the box in the Ammo box, like Hornady 140gr ELD Match. A box is one group and one string: measure the target and bring in the speeds the same way as any load. Save each box as its own load, and Compare Loads ranks boxes against each other, or against your handloads.

Using a chronograph or target app that isn't listed? Send a sample export to support@loadscope.app and we'll add it — the file readers are pluggable, so a new one is quick.